Java String Case Study: Employee ID & Email Validation System

Scenario

Your company, ABC Technologies, is developing an Employee Management System. During employee registration, HR has noticed inconsistent employee names, invalid email addresses, incorrect employee IDs, weak passwords, duplicate email addresses, and the need to generate usernames automatically. As a Java developer, use String methods to validate and process employee information.

Sample Input

Employee Name: ' John Smith '
Email: John.Smith@ABC.com
Employee ID: EMP-1023
Password: John@123

Tasks

1. Trim employee names.
2. Convert emails to lowercase.
3. Validate email format.
4. Validate employee ID (EMP-####).
5. Generate usernames (firstname.lastname).
6. Count words in names.
7. Check password strength.
8. Mask email addresses.
9. Detect duplicate emails.
10. Extract email domain.

Expected Output

Display a registration report showing cleaned name, normalized email, validation status, generated username, password strength, email domain, and employee code.

Java String Concepts

trim(), length(), charAt(), substring(), split(), contains(), startsWith(), endsWith(), indexOf(), lastIndexOf(), replace(), replaceAll(), equals(), equalsIgnoreCase(), toLowerCase(), toUpperCase(), matches().

Extension Activity

Read employee records from a CSV file, validate each record, export cleaned data to a new CSV, and generate summary statistics.